

STAT451: Applied Statistics for Engineering and Sciences

Practice Final Exams (Based on 2024-2025 Format)

Extracted from: Devore, J. L. (2015). *Probability and Statistics for Engineering and the Sciences*

General Instructions

- Duration: 90 minutes.
 - You are allowed one A4-sized formula sheet (double-sided).
 - Round all final calculations to four decimal places.
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Practice Test 1

Question 1 (2.0 points)

A particular brand of dishwasher soap is sold in three sizes: 25 oz, 40 oz, and 65 oz. Twenty percent of all purchasers select a 25-oz box, 50% select a 40-oz box, and the remaining 30% choose a 65-oz box. Let $X_1, X_2, \dots, X_{50}$ be a random sample of 50 independent purchases.

- Determine the expected value and variance of the size of a single box.
- Use the Central Limit Theorem to approximate the probability that the total weight of the 50 boxes exceeds 2250 oz.

Source: Devore, Page 229, Exercise 37.

Question 2 (1.5 points)

The following observations are on drill lifetime (number of holes) for a brass alloy:

161, 168, 184, 206, 248, 263, 289, 322, 388, 513

- Calculate the sample mean and the 10% trimmed mean.
- Calculate the sample variance and sample standard deviation.

Source: Devore, Page 46, Exercise 27.

Question 3 (2.0 points)

A sample of 50 lenses for eyeglasses yields a sample mean thickness of 3.05 mm and a sample standard deviation of 0.34 mm.

- Calculate a 95% confidence interval for the true average thickness μ .
- If the desired width of the 95% interval is 0.1 mm, what sample size is required?

Source: Devore, Page 357, Exercise 57.

Question 4 (2.0 points)

In an experiment to compare the bond strength of two adhesives, five bondings were made with each type.

Adhesive 1: 229, 286, 245, 299, 250

Adhesive 2: 213, 179, 163, 247, 225

Assume normal distributions with equal variances. Test $H_0 : \mu_1 = \mu_2$ against $H_a : \mu_1 > \mu_2$ at $\alpha = 0.05$.

Source: Devore, Page 666, Exercise 11.

Question 5 (2.5 points)

Observations on x = cation exchange capacity and y = specific surface area for 20 soils resulted in:

$\sum x = 1762$, $\sum y = 5480$, $\sum x^2 = 164012$, $\sum y^2 = 1654200$, $\sum xy = 498300$.

(a) Determine the equation of the least squares regression line.

(b) Calculate and interpret the coefficient of determination R^2 .

Source: Devore, Page 536, Exercise 62.

Practice Test 2

Question 1 (2.0 points)

A company knows that 10% of its goblets have cosmetic flaws.

- (a) Among 15 randomly selected goblets, what is the probability that at least 4 are "seconds"?
- (b) In a sample of 200 goblets, use the normal approximation to find the probability that between 15 and 25 (inclusive) are seconds.

Source: Devore, Page 123, Exercise 49.

Question 2 (2.0 points)

In a sample of 1000 consumers, 250 said they never sent in a rebate claim.

- (a) Calculate a 99% confidence interval for the true proportion p of consumers who never apply for a rebate.
- (b) What sample size is needed to ensure the margin of error is at most 0.02 with 95% confidence?

Source: Devore, Page 294, Exercise 21.

Question 3 (1.5 points)

A mixture of fuel ash and cement should have a compressive strength $\mu > 1300 \text{ KN/m}^2$. For $n = 10$ specimens, $\bar{x} = 1340$. Assume normality with $\sigma = 60$. Test $H_0 : \mu = 1300$ vs $H_a : \mu > 1300$ at $\alpha = 0.01$.

Source: Devore, Page 326, Exercise 12.

Question 4 (2.0 points)

Computer retrieval times were measured for 13 professionals using two methods: Slide (x) and Digital (y). The differences ($d = x - y$) had $\bar{d} = 20.5$ and $s_d = 11.96$. Test if the Digital method is faster at $\alpha = 0.05$. *Source: Devore, Page 389, Exercise 38.*

Question 5 (2.5 points)

Data on x = duration of a tantrum and y = anger intensity was collected. Given $n = 12$, $\sum x = 150$, $\sum y = 85$, $S_{xx} = 450$, $S_{yy} = 210$, $S_{xy} = 280$.

- (a) Find the regression line $y = \hat{\beta}_0 + \hat{\beta}_1 x$.
- (b) Predict the intensity for a 15-minute tantrum and discuss if this is appropriate.

Source: Devore, Page 45, Exercise 23.

Practice Test 3

Question 1 (2.0 points)

The processing time X for garbage trucks is normally distributed with $\mu = 13$ min and $\sigma = 4$ min.

- (a) What is the probability that a single truck takes more than 20 min?
- (b) For a sample of $n = 16$ trucks, what is the probability that the sample mean time $\bar{X}$ is between 12 and 15 min?

Source: Devore, Page 245, Exercise 87.

Question 2 (2.0 points)

A package-delivery service claims 90% of packages are delivered by noon. In a sample of 225 packages, 180 were delivered by noon. Does this suggest the true proportion is less than 0.90? Test at $\alpha = 0.01$.

Source: Devore, Page 349, Example 8.14.

Question 3 (2.0 points)

Mean headability ratings for 30 specimens of aluminum killed steel ($\bar{x}_1 = 7.09, s_1 = 1.19$) and 30 specimens of silicon killed steel ($\bar{x}_2 = 6.43, s_2 = 1.08$) were compared. Use a t -test to determine if the means differ at $\alpha = 0.05$. *Source: Devore, Page 405, Exercise 75.*

Question 4 (1.5 points)

The following are CeO_2 particle sizes: 3.0, 3.4, 3.8, 3.9, 4.0, 4.2, 4.5, 4.8, 5.1, 7.2.

- (a) Identify any outliers using the IQR method.
- (b) Compare the mean and median; what does this suggest about the skewness?

Source: Devore, Page 43, Exercise 12.

Question 5 (2.5 points)

A study of $x =$ stiffness and $y =$ thickness for flame-retardant fabrics gave $n = 6, \sum x = 112.7, \sum y = 2.91, \sum x^2 = 3012, \sum y^2 = 1.68, \sum xy = 65.4$.

- (a) Calculate the sample correlation coefficient r .
- (b) Test $H_0 : \rho = 0$ vs $H_a : \rho \neq 0$ at $\alpha = 0.05$.

Source: Devore, Page 536, Exercise 63.